

(b) NWM pretraining

Which pretraining signal transfers best?

Base data · GT navigation actions

RECON · SCAND · TartanDrive · HuRoN

NavAnywhere v1

13 datasets · 322 h

NavAnywhere v2 · scale extension

v1 + Ego4D + GO · 1200+ h

MODEL	CONDITIONING SIGNAL	PRETRAINING	BASE-DATA TRAINING
nwm-real	$a \rightarrow E_a$	None	Train from scratch
nwm-timept	Time Δt only	NA-v1 $\rightarrow$ NWM	Reset fine-tuning
nwm-geopt	VGGT $\rightarrow \tilde{a} \rightarrow E_{\text{geo}}$	NA-v1 $\rightarrow$ NWM	Reset fine-tuning
nwm-latentpt	$E_{\text{LAM}} \rightarrow z \rightarrow E_z$	NA-v1 $\rightarrow$ NWM	3 strategies in (c)

TimePT: no action labels. GeoPT / LatentPT: proxy actions from video. c denotes action conditioning.

Shared NWM prediction

